

Rest of the solution to exercise 10.2.:

We wanted to show that φ is bijective. We have already seen that φ is injective. Now we have to prove that it is also surjective.

Note: The domain and co-domain of φ are infinite. Thus, it will not suffice to argue that they have the same cardinality and that φ is injective.

Let (x, y) be an arbitrary element of $\mathbb{Q} \times \mathbb{Q}$. Now choose f as the polynomial:

$$f(t) := \frac{x-y}{2} \cdot t + \frac{x+y}{2}$$

It is easy to see that:

$$\varphi([f]_g) = (f(1), f(-1)) = \left(\frac{x-y}{2} + \frac{x+y}{2}, \frac{y-x}{2} + \frac{x+y}{2} \right) = (x, y)$$

This implies that φ is surjective.

Regarding the zero divisors and invertible elements of S : Since φ is a ring homomorphism, we have for every $[f]_g \in S$:

$$\begin{aligned} [f]_g &\text{ is a zero divisor in } S. \\ \Leftrightarrow \varphi([f]_g) = (f(1), f(-1)) &\text{ is a zero divisor in } \mathbb{Q} \times \mathbb{Q}. \\ \Leftrightarrow f(1) = 0 &\text{ or } f(-1) = 0. \end{aligned}$$

So, the zero divisors of S are exactly those $[f]_g \in S$ with $f(1) = 0$ or $f(-1) = 0$.

Analog:

$$\begin{aligned} [f]_g &\text{ is invertible in } S. \\ \Leftrightarrow \varphi([f]_g) = (f(1), f(-1)) &\text{ is invertible in } \mathbb{Q} \times \mathbb{Q}. \\ \Leftrightarrow f(1) \neq 0 &\text{ and } f(-1) \neq 0. \end{aligned}$$

So, the invertible elements of S are exactly those $[f]_g \in S$ with $f(1) \neq 0$ and $f(-1) \neq 0$.